Bayesian Probabilistic Selection index

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14 November 2025

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1 Introduction

This workflow will help you in the Bayesian probabilistic selection index (BPSI) usage. To apply this tool for your crop, it's recommended pay attention to each section and run carrefully the codes. Any questions? Contact us by ProbBreed or [e-mail][“mailto:josetchagas@usp.br”].

1.1 Pan-African Soybean Trials dataset

Cultivar recommendation is a critical stage in plant breeding programs, and selecting superior genotypes for multiple traits remains a challenge due to the genotypes × environment (G × E) interaction.

To address this, we analyzed multi-environment trials data from Pan-African Trials, evaluating 62 soybean genotypes across 19 environments (Araújo et al. 2025).

Our methodological approach integrates Bayesian probabilistic framework to estimate genotypes risk recommendation to each trait and select genotypes according to a multitrait ideotype (Chagas et al. 2025). The Bayesian probabilistic selection index (BPSI) select the 10% genotypes with higher probability of superior performance across environments to Grain Yield, Plant Height and Number of Days to Maturity.

1.2 Bayesian probabilistic selection index (BPSI)

The Bayesian probabilistic selection index enables design ideotype by increase or decrease and weighting traits. Along with the probabilities of superior performance across environments, it accounts for the distance between the worst genotype performance and candidate genotypes for each trait.

$$BPSI_i = \sum_{m=1}^{t} \frac{\gamma_{pt} - \gamma_{it}}{(1/\lambda_t)}$$

where γ_p is the probability of superior performance of the worst genotype for the trait m, γ is the probability of superior performance of genotype i for trait m, t is the total number of traits evaluated, $(m = 1, 2, ..., t)$, and λ is the weight for each trait t.

2 BPSI example at soybean dataset

In order to provide a step-by-step guide, we applied

2.1 Bayesian Model

The Bayesian model are available at ProbBreed package (Chaves et al. 2024). We used the entry mean model.

$$y_{jk} = \mu + l_k + g_j + \varepsilon_{jk}$$

where the y_{jk} is the phenotypic record of the genotype j^{th} in the k^{th} location, μ is the intercept, l_k is the main effects of the location, g_j is the main effect of the genotype, and ϵ_{jk} is the residual effect.

```
library(ProbBreed)

met_df=read.csv("https://raw.githubusercontent.com/tiagobchagas/BPSI/refs/heads/main/Data/blues_long.cs

head(met_df)
```

```
##   env gen       PH       GY NDM
## 1 E01 G02 77.56172 3690.918  NA
## 2 E01 G05 81.16229 1004.449  NA
## 3 E01 G09 62.17745 3062.170  NA
## 4 E01 G11 33.37286 1804.674  NA
## 5 E01 G14 52.03038 2719.217  NA
## 6 E01 G21 97.85586 1747.515  NA
```

```
mod = bayes_met(data = met_df,
                gen = "gen",
                loc = "env",
                repl = NULL,
                trait = "PH",
                reg = NULL,
                year = NULL,
                res.het = T,
                iter =40, cores = 4, chain = 4) #recommended run at least 4k iterations


mod2 = bayes_met(data = met_df,
                 gen = "gen",
                 loc = "env",
                 repl = NULL,
                 trait = "GY",
                 reg = NULL,
                 year = NULL,
                 res.het = T,
                 iter = 40, cores = 4, chain = 4) #recommended run at least 4k iterations

mod3 = bayes_met(data = met_df,
                 gen = "gen",
                 loc = "env",
```

```
                 repl =  NULL,
                 trait = "NDM",
                 reg = NULL,
                 year = NULL,
                 res.het = T,
                 iter = 40, cores = 4, chain = 4) #recommended run at least 4k iterations
```

2.2 Extracting the probability of superior performance

```
models=list(mod,mod2,mod3)
names(models) <- c("PH","GY","NDM")
inc=c(FALSE,TRUE,FALSE)
names(inc) <- names(models)

results <- lapply(names(models), function(model_name) {
  x <- models[[model_name]]  # actual model object

  outs <- extr_outs(model = x,
                    probs = c(0.05, 0.95),
                    verbose = TRUE)

  a <- prob_sup(extr = outs,
                int = .2,
                increase = inc[[model_name]],  # ← now model_name is a character!
                save.df = FALSE,
                verbose = TRUE)

  return(a)
})
```

```
##
## Divergences:
##
## Tree depth:
##
## Energy:
##
## Divergences:
##
## Tree depth:
##
## Energy:
##
## Divergences:
##
## Tree depth:
##
## Energy:
```

```r
names(results) <- names(models)
```

2.3 Running the BPSI

```r
source("Data/bpsi.R")



bpsi=BPSI(problist=results,
          increase = c(FALSE,TRUE,FALSE),
          int = 0.1,
          lambda =c(1,2,1),
          save.df = F)
```

```
## -> BPSI selected 10% of genotypes
```

```
## Process completed!
```

2.4 BPSI Ranks

The selected genotypes (G16, G14, G11, G50, G30, G43) have superior performance for the three traits evaluated Grain Yield (GY), Plant Height (PH) and Number of Days to Maturity (NDM).

```r
plot(bpsi,category = "Ranks")
```

```
## Loading required package: ggplot2
```

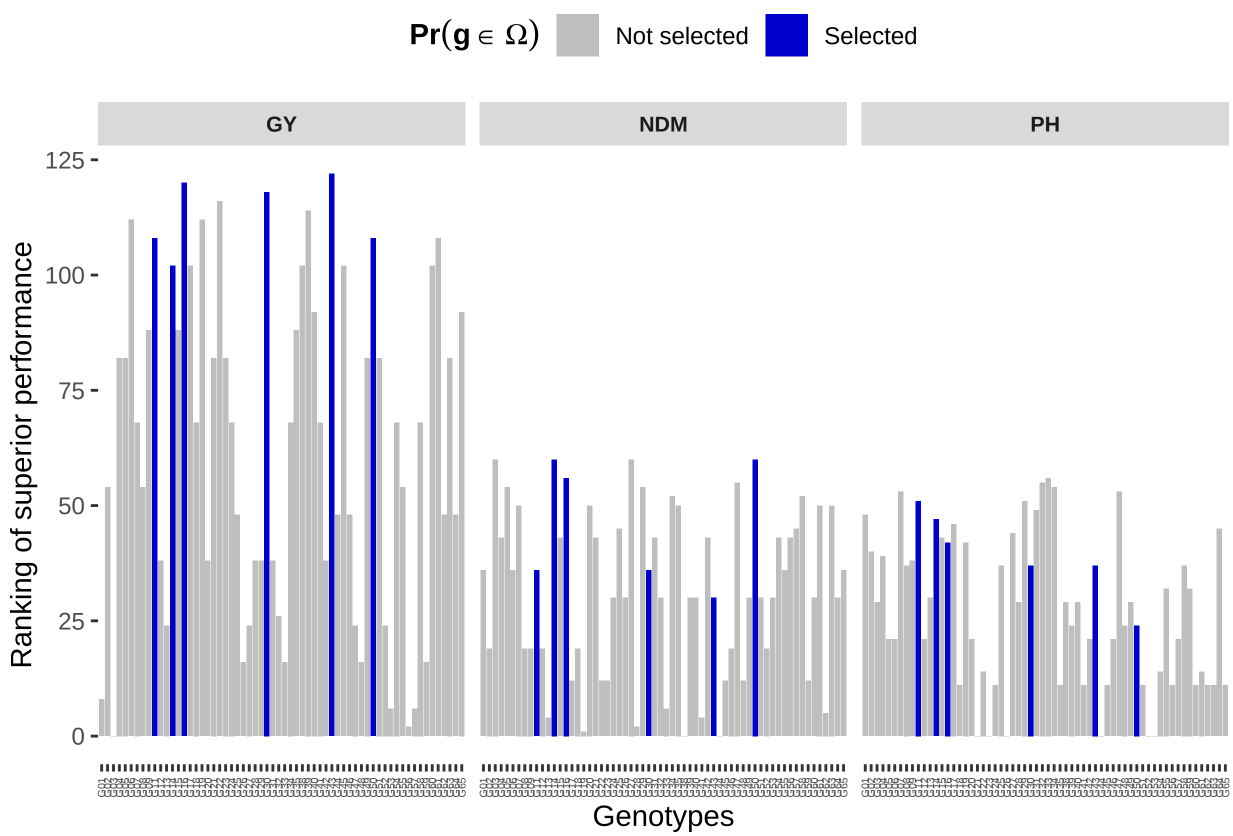

Figure 1: Rank of probability of superior performance of soybean Pan-African Trials across locations in Kenya

2.5 BPSI plot

The selected genotypes (G16, G14, G11, G50, G30, G43) have minor risk of cultivar recomendation to the multitrait ideotype across the Kenya environments.

```
plot(bpsi,category = "BPSI")
```

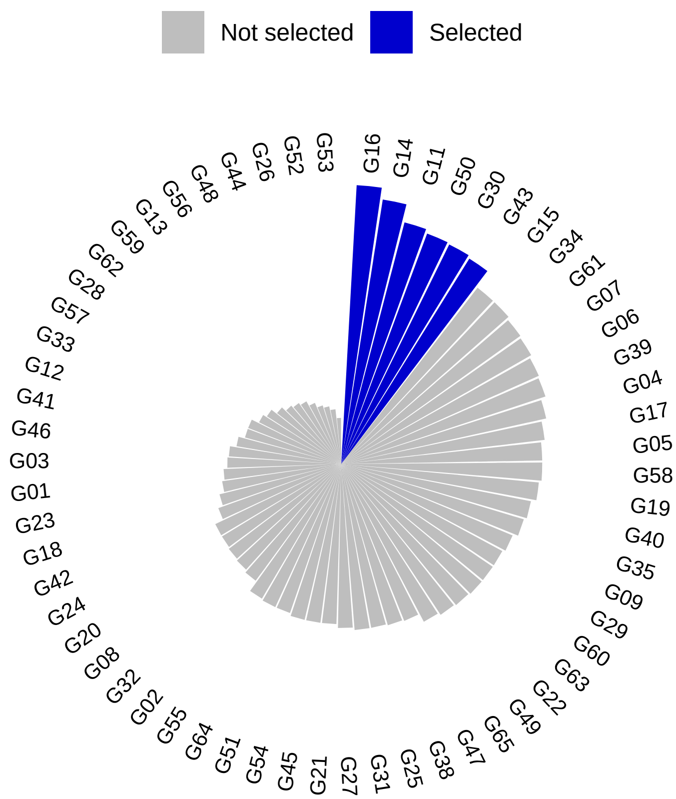

Figure 2: Bayesian probability selection index of soybean genotypes from Pan-African Trials across locations in Kenya

2.6 Probabilities of superior performance

The probability os superior performance across environments show the risk recommendation of the soybean genotypes being in the top 20% to Grain Yield (GY) and the bottom 20% for Plant Height (PH) and Number of Days to Maturity (NDM).

```
df=bpsi$df
df |> kableExtra::kable()
```

	PH	GY	NDM
G01	0.6625	0.1000	0.2000
G02	0.2625	0.1875	0.1750
G03	0.0750	0.0375	0.3000
G04	0.2000	0.2125	0.2125
G05	0.0375	0.2125	0.2625
G06	0.0375	0.2875	0.2000
G07	0.8500	0.2000	0.2375
G08	0.1250	0.1875	0.1750

	PH	GY	NDM
G09	0.1750	0.2250	0.1750
G11	0.8000	0.2750	0.2000
G12	0.0375	0.1500	0.1750
G13	0.0875	0.1250	0.1250
G14	0.6500	0.2625	0.3000
G15	0.3500	0.2250	0.2125
G16	0.3375	0.3750	0.2875
G17	0.5375	0.2625	0.1625
G18	0.0125	0.2000	0.1750
G19	0.3375	0.2875	0.1000
G20	0.0375	0.1500	0.2375
G21	0.0000	0.2125	0.2125
G22	0.0250	0.3125	0.1625
G23	0.0000	0.2125	0.1625
G24	0.0125	0.2000	0.1875
G25	0.1250	0.1625	0.2250
G26	0.0000	0.1125	0.1875
G27	0.3625	0.1250	0.3000
G28	0.0750	0.1500	0.1125
G29	0.8000	0.1500	0.2625
G30	0.1250	0.3625	0.2000
G31	0.7500	0.1500	0.2125
G32	0.9000	0.1375	0.1875
G33	0.9250	0.1125	0.1500
G34	0.8625	0.2000	0.2500
G35	0.0125	0.2250	0.2375
G38	0.0750	0.2625	0.0875
G39	0.0625	0.3000	0.1875
G40	0.0750	0.2375	0.1875
G41	0.0125	0.2000	0.1250
G42	0.0375	0.1500	0.2125
G43	0.1250	0.3875	0.1875
G44	0.0000	0.1625	0.0875
G45	0.0125	0.2625	0.1625
G46	0.0375	0.1625	0.1750
G47	0.8500	0.1250	0.2750
G48	0.0625	0.1125	0.1625
G49	0.0750	0.2125	0.1875
G50	0.0625	0.2750	0.3000
G51	0.0125	0.2125	0.1875
G52	0.0000	0.1250	0.1750
G53	0.0000	0.0875	0.1875
G54	0.0250	0.2000	0.2125
G55	0.1125	0.1875	0.2000
G56	0.0125	0.0750	0.2125
G57	0.0375	0.0875	0.2250
G58	0.1250	0.2000	0.2500
G59	0.1125	0.1125	0.1625
G60	0.0125	0.2625	0.1875
G61	0.0250	0.2750	0.2375
G62	0.0125	0.1625	0.1375
G63	0.0125	0.2125	0.2375

	PH	GY	NDM
G64	0.4125	0.1625	0.1875
G65	0.0125	0.2375	0.2000

2.7 Genotypes selected

The genotypes by BPSI selected were G16, G14, G11, G50, G30, G43. It have the greater distance to the worst genotype to Grain Yield (GY), Plant Height (PH) and Number of Days to Maturity (NDM). The genotype selected are the 10% multitrait top performance across enviroments.

```
df=bpsi$BPSI

df |> kableExtra::kable()
```

	PH	GY	NDM	bpsi	sel	gen
G16	42	120	56	218	Selected	G16
G14	47	102	60	209	Selected	G14
G11	51	108	36	195	Selected	G11
G50	24	108	60	192	Selected	G50
G30	37	118	36	191	Selected	G30
G43	37	122	30	189	Selected	G43
G15	43	88	43	174	Not_Selected	G15
G34	54	68	52	174	Not_Selected	G34
G61	14	108	50	172	Not_Selected	G61
G07	53	68	50	171	Not_Selected	G07
G06	21	112	36	169	Not_Selected	G06
G39	24	114	30	168	Not_Selected	G39
G04	39	82	43	164	Not_Selected	G04
G17	46	102	12	160	Not_Selected	G17
G05	21	82	54	157	Not_Selected	G05
G58	37	68	52	157	Not_Selected	G58
G19	42	112	1	155	Not_Selected	G19
G40	29	92	30	151	Not_Selected	G40
G35	11	88	50	149	Not_Selected	G35
G09	38	88	19	145	Not_Selected	G09
G29	51	38	54	143	Not_Selected	G29
G60	11	102	30	143	Not_Selected	G60
G63	11	82	50	143	Not_Selected	G63
G22	14	116	12	142	Not_Selected	G22
G49	29	82	30	141	Not_Selected	G49
G65	11	92	36	139	Not_Selected	G65
G47	53	24	55	132	Not_Selected	G47
G38	29	102	0	131	Not_Selected	G38
G25	37	48	45	130	Not_Selected	G25
G31	49	38	43	130	Not_Selected	G31
G27	44	24	60	128	Not_Selected	G27
G21	0	82	43	125	Not_Selected	G21
G45	11	102	12	125	Not_Selected	G45
G54	14	68	43	125	Not_Selected	G54
G51	11	82	30	123	Not_Selected	G51
G64	45	48	30	123	Not_Selected	G64
G55	32	54	36	122	Not_Selected	G55
G02	40	54	19	113	Not_Selected	G02

	PH	GY	NDM	bpsi	sel	gen
G32	55	26	30	111	Not_Selected	G32
G08	37	54	19	110	Not_Selected	G08
G20	21	38	50	109	Not_Selected	G20
G24	11	68	30	109	Not_Selected	G24
G42	21	38	43	102	Not_Selected	G42
G18	11	68	19	98	Not_Selected	G18
G23	0	82	12	94	Not_Selected	G23
G01	48	8	36	92	Not_Selected	G01
G03	29	0	60	89	Not_Selected	G03
G46	21	48	19	88	Not_Selected	G46
G41	11	68	4	83	Not_Selected	G41
G12	21	38	19	78	Not_Selected	G12
G33	56	16	6	78	Not_Selected	G33
G57	21	6	45	72	Not_Selected	G57
G28	29	38	2	69	Not_Selected	G28
G62	11	48	5	64	Not_Selected	G62
G59	32	16	12	60	Not_Selected	G59
G13	30	24	4	58	Not_Selected	G13
G56	11	2	43	56	Not_Selected	G56
G48	24	16	12	52	Not_Selected	G48
G44	0	48	0	48	Not_Selected	G44
G26	0	16	30	46	Not_Selected	G26
G52	0	24	19	43	Not_Selected	G52
G53	0	6	30	36	Not_Selected	G53

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